



Marco Antonio Adame Rodríguez

IoT & Embedded Systems Engineer

Digital Signal Processing · Scientific Software

Mexico (GMT-6) — Available for remote work with US and European overlap

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200 Hz

Synchronized sampling

2048

Points per FFT window

0.98 Hz

Frequency resolution

3

Published papers

Profile

Computer Engineer and graduate of the MSc in Engineering for Innovation and Technological Development, with a terminal specialization in Information and Communication Technologies (UAGro). All coursework completed with a 9.42/10 GPA; degree conferral in progress, expected November 2026. I build end-to-end IoT systems: MEMS sensor, embedded firmware, real-time acquisition, Linux backend, cloud storage and visualization dashboard. My graduate work is a low-cost structural health monitoring system I designed from scratch and validated in the field against commercial seismic instrumentation.

Research & development experience

Lead Researcher and Developer — SHM-IoT System

Aug 2024 – Present

UAGro · MIIDT graduate program

End-to-end design and implementation of a low-cost structural health monitoring solution for civil works, field-validated against commercial GEA and P-Alert equipment.

Acquisition, firmware and hardware

- Integrated acquisition nodes on Arduino Nano RP2040 Connect and Raspberry Pi 4 with MPU9250/MPU6050, LSM6DSOX and ADXL335 MEMS sensors over I2C, SPI and UART.
- Implemented synchronized dual-core acquisition at 200 Hz, satisfying the Nyquist criterion for the structural frequency range of interest.
- Validated the system on a biaxial shaking table, university buildings, a botanical garden, a high school campus and lightweight experimental structures.

Signal processing and scientific software

- Built the DSP pipeline: RMS, FFT, power spectral density (Welch), dominant frequency, coherence, phase analysis and cross-correlation.
- Developed Python and PyQt5 applications for acquisition, live monitoring and analysis, with session control and automatic crash recovery.
- Automated the generation of technical reports and comparative plots against reference equipment.

Backend, cloud and infrastructure

- Stood up a centralized multi-site receiver over HTTP/HTTPS with local buffering that prevents data loss during connectivity outages.
- Developed REST APIs in FastAPI for ingesting, persisting and querying ESP32 node telemetry.
- Administered continuously running Ubuntu servers: systemd, SSH/VNC, Docker, backup and synchronization.
- Published web dashboards to visualize, filter and download data from both custom nodes and commercial sensors.

Publications & talks

[1]

Ambient Vibration Measurement System on a Biaxial Vibrating Table
Adame Rodríguez, M. A., Alcaraz Vázquez, M., Martínez, E. T., García Guevara, D. S., Alonso Silverio, G. A. (2026). Revista Vínculos, 22(1). **Lead author**

[2]

Feature-importance-based attribute selection for dengue case classification in Mexico
Alcaraz Vázquez, M., Adame Rodríguez, M. A., García Guevara, D. S., Martínez, E. T., Alonso Silverio, G. A. (2025). Revista Vínculos, 22(2). **Co-author**

[3]

IoT-based biological sample shaker with temperature control
García Guevara, D. S., Martínez, E. T., Alcaraz Vázquez, M., Adame Rodríguez, M. A., Alonso Silverio, G. A. (2025). Revista Vínculos, 22(2). **Co-author**

Conference talk. "Ambient Vibration Measurement System on a Biaxial Vibrating Table". 15th International Computing Conference (CICOM 2025), Tunja, Colombia, October 2025.

Technical skills

Languages	Python, C/C++ (embedded), JavaScript, SQL, PHP, Bash
Data & signals	NumPy, Pandas, SciPy, Matplotlib, FFT, PSD/Welch, RMS, Digital filters, Time series
AI & machine learning	Machine Learning, Deep Learning, Computer Vision, Big Data, Feature selection
IoT & hardware	Raspberry Pi, ESP32, Arduino Nano RP2040, MEMS sensors, I2C, SPI, UART
Backend & cloud	FastAPI, REST APIs, Supabase, PostgreSQL, Firebase
Systems & DevOps	Linux / Ubuntu Server, Docker, systemd, SSH, Git, Vercel

Projects

SHM-IoT MIIDT — Low-cost structural monitoring
Python · C++ · Raspberry Pi · PyQt5

Full architecture for acquisition, storage, analysis and visualization of accelerations in civil structures, field-tested against commercial sensors.

Geo-referenced SHM monitoring network
JavaScript · Vercel

Web dashboard showing location and status of distributed monitoring nodes, continuously deployed and public.
[github.com/Ryukert/SHM_GPS_NETWORK_DASH_elite](#)

Supabase sensor dashboard
JavaScript · Supabase · PostgreSQL

Web application for submitting, querying and visualizing measurements stored in the cloud.
[github.com/Ryukert/MIIDT_SUPABASE_DASH](#)

ESP32 telemetry APIs
Python · FastAPI

REST services for receiving, persisting and querying acceleration data, plus scientific plotting tools.
[github.com/Ryukert/APIAceleraciones](#)

Computing areas covered

All 69 courses passed across both degrees, organised by computing area. "Applied" marks the areas I also back with real work — systems in production, peer-reviewed publications or field validation; "Coursework" means academic coverage without an associated public project.

- **Programming, algorithms and data structures**
Python and C/C++ daily: embedded firmware, processing pipelines and desktop tooling.
Programming Fundamentals (9) Object-Oriented Programming I (10) Object-Oriented Programming II (8) Advanced Programming (8) Data Structures I (10) Data Structures II (9) Computational Logic (8)

- **Software engineering and system design**
PyQt5 applications with session control, data export and automatic crash recovery.
Systems Analysis and Design (10) Software Engineering (8)

- **Compilers and language theory**
Compilers (9) Translators and Interpreters (8)

- **Operating systems and server administration**
Continuously running Ubuntu servers: systemd, SSH/VNC, backups and synchronization.
Operating Systems I (10) Operating Systems II (8) Server Architecture (10)

- **Networking and communications**
Multi-site monitoring with nodes on separate networks and a centralized HTTP/HTTPS receiver.
Routing Fundamentals (10) Network Fundamentals (9) Structured Cabling Systems (9) Communications Fundamentals (7)

- **Security, auditing and computer forensics**
Complemented by the Cisco Cybersecurity Essentials and CCNAv7 certifications.
IT Resource Auditing (9) Network Security (8) Computer Forensics (8)

- **Databases and persistence**
PostgreSQL and Supabase as multi-origin cloud storage for telemetry.
Databases II (9) Databases I (7)

- **Computer architecture and digital systems**
Synchronized acquisition on a dual-core RP2040 architecture.
Digital Systems (10) Computer Organization (10) Microprocessors (8)

- **Embedded systems, IoT and electronics**
The core of my work: RP2040 and Raspberry Pi nodes with MEMS sensors over I2C, SPI and UART.
Selected Topics in Embedded Systems (10) Electricity and Magnetism (10) Microcontrollers (9) IoT Systems Design (9) Electrical Circuits (9) Fundamentals of Electronic Systems (9) Electronics (7)

- **Artificial intelligence and machine learning**
Co-author of the paper on feature-importance-based attribute selection for dengue case classification.
Machine Learning (10) Deep Learning (10) Data Analysis and Big Data (10) Artificial Intelligence Fundamentals (7)

- **Signal, vision and image processing**
Custom DSP pipeline: FFT, Welch PSD, RMS, coherence, phase and cross-correlation (published and peer-reviewed).
Computer Vision (10) Digital Image Processing (7)

- **Mathematics, statistics and numerical methods**
The basis of the spectral analysis: Nyquist criterion, frequency resolution and statistical spectrum estimation.
Differential and Integral Calculus (8) Probability and Statistics (8) Numerical Methods (8) Analytic Geometry (8) Basic Chemistry (8) Vector Calculus (7) Differential Equations (7) Algebra (7) General Physics (7)

- **Operations research and optimization**
Operations Research (8)

- **Interfaces, visualization and application development**
Web dashboards deployed on Vercel and desktop interfaces for live monitoring.
Mobile Application Development (10) Human-Computer Interaction (9)

- **Information technology and innovation**
A low-cost alternative to commercial seismic instrumentation, validated against GEA and P-Alert.
Managing Information and Communication Technologies (10) Information and Communication Technologies (10) Sustainable Innovation and Technological Development (10)

- **Research and scientific communication**
Three papers in Revista Vínculos and a talk at CICOM 2025 (Tunja, Colombia).
Communication Skills (10) Logical, Heuristic and Creative Thinking (10) Research Seminar I (9) Graduate Research III (9) Research Seminar II (8) Graduate Research I (8) Graduate Research II (8) Graduation Project (completed)

- **General education and practicum**
950 certified hours across social service and a professional internship.
Entrepreneurship Workshop (10) Contemporary World Analysis (10) English I (7) English II (7) Professional Residency (completed)
Professional Internship (completed) Social Service (completed)

Education

MSc in Engineering for Innovation and Technological Development
Universidad Autónoma de Guerrero

Aug 2024 – 2026

GPA 9.42 / 10 All coursework completed No failed courses

Terminal specialization: Information and Communication Technologies (2023 curriculum). All coursework completed; degree conferral in progress, expected November 2026. Thesis: "Development and implementation of a low-cost IoT system for structural health monitoring in civil works".

Grades on a 0–10 scale. The Professional Residency and Graduation Project are pass/fail and excluded from the GPA.

Artificial intelligence and data	Machine Learning (10) Deep Learning (10) Computer Vision (10) Data Analysis and Big Data (10)
Embedded systems and IoT	Selected Topics in Embedded Systems (10) IoT Systems Design (9) Fundamentals of Electronic Systems (9)
Programme core	Information and Communication Technologies (10) Sustainable Innovation and Technological Development (10) Graduate Research I (8) Graduate Research II (8) Graduate Research III (9) Professional Residency (completed) Graduation Project (completed)

BSc in Computer Engineering
Universidad Autónoma de Guerrero

2018 – 2023

GPA 8.56 / 10 55 courses passed

School of Engineering, Chilpancingo. Degree awarded 2023. Federal professional license no. 13426767.

Grades on a 0–10 scale; 6 is the minimum passing grade.

Programming and software engineering	Programming Fundamentals (9) Object-Oriented Programming I (10) Object-Oriented Programming II (8) Advanced Programming (8) Data Structures I (10) Data Structures II (9) Computational Logic (8) Systems Analysis and Design (10) Software Engineering (8) Translators and Interpreters (8) Compilers (9) Mobile Application Development (10) Human-Computer Interaction (9)
Hardware, electronics and architecture	Digital Systems (10) Computer Organization (10) Server Architecture (10) Microcontrollers (9) Microprocessors (8) Electrical Circuits (9) Electricity and Magnetism (10) Electronics (7)
Networking, operating systems and security	Routing Fundamentals (10) Operating Systems I (10) Operating Systems II (8) Network Fundamentals (9) Structured Cabling Systems (9) IT Resource Auditing (9) Network Security (8) Computer Forensics (8) Communications Fundamentals (7)
Data, artificial intelligence and signals	Databases II (9) Databases I (7) Artificial Intelligence Fundamentals (7) Digital Image Processing (7)
Mathematics and basic sciences	Differential and Integral Calculus (8) Probability and Statistics (8) Numerical Methods (8) Analytic Geometry (8) Operations Research (8) Basic Chemistry (8) Vector Calculus (7) Differential Equations (7) Algebra (7) General Physics (7)
Research and general education	Research Seminar I (9) Research Seminar II (8) Entrepreneurship Workshop (10) Managing Information and Communication Technologies (10) Logical, Heuristic and Creative Thinking (10) Communication Skills (10) Contemporary World Analysis (10) English I (7) English II (7)
Practicum	Professional Internship (completed) Social Service (completed)

Additional experience

Jóvenes Construyendo el Futuro — Publishing department
Ministry of Labour (Mexico)

Jun 2023 – Jun 2024

12-month vocational training programme in digital production and printing.

Professional internship — IT area
35th Military Zone (Mexican Army)

Aug 2022 – Jan 2023

450 certified hours. Computer equipment support and maintenance in an institutional environment.

Social service — Computing area

Feb 2022 – Jul 2022

Ignacio Manuel Altamirano Secondary School

500 certified hours. Technical support and community-sector educational support.

Certifications

Data Science Essentials with Python — Cisco / UAGro (Dec 2025)

Cybersecurity Essentials — Cisco (Mar 2022)

NDG Linux Unhatched — Cisco (May 2022)

CCNAv7: Introduction to Networks — Cisco (Jun 2021)

Languages

Spanish: native.

English: intermediate (B1), TOEFL ITP 470 (2025). Fluent reading of technical documentation.

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